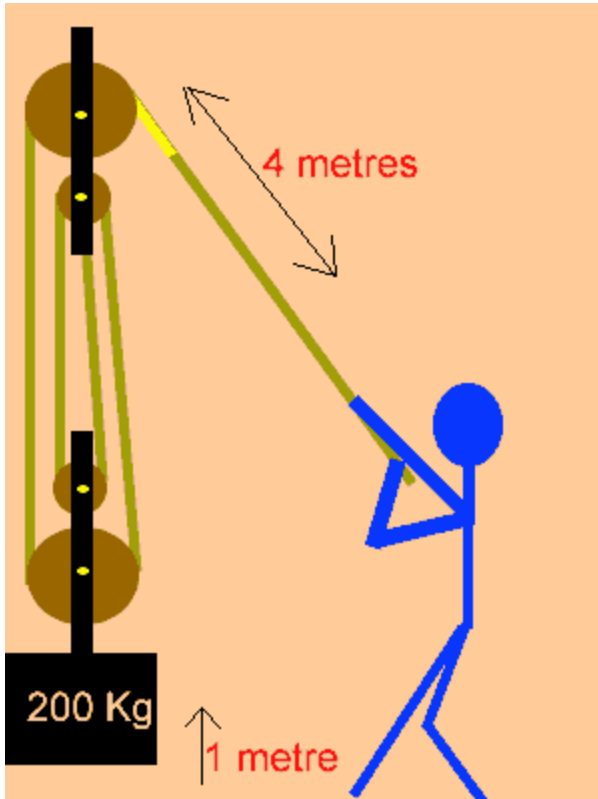


popquiz inclined plane, equilibrium

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:09pm

Quiz Instructions



Question 1 10 pts

A 50kg crate is **sliding down** a frictionless slope (incline plane). The mass of the crate is m . The angle of the slope is 40 degrees (see slide with skier). Take the x-axis along the slope (surface).

The x-component of the weight of the crate is: (take $g=10\text{m/s/s}$ and round to nearest one

remember I told you to memorize it. the x-component of the weight along an inclined is always...

see my slide about the skier.



321N



500N



383N



32N



Question 2 10 pts

Building on the previous question.

The normal force on the crate is: ($g=10\text{m/s/s}$ round to nearest one). normal is along the +yaxis. its balancing out the y-component of the weight. I told you to memorize thats the y-component of the weight is for inclined plane.



383N



0N



321N

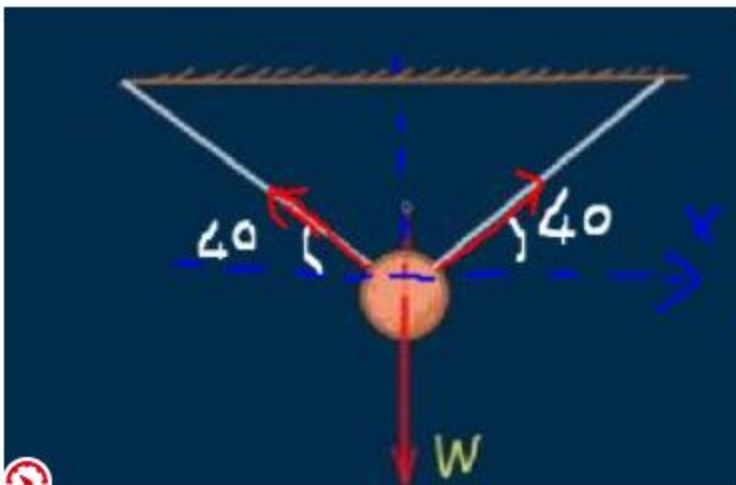


500N



Question 3 10 pts

The weight is 500N. T1 is pulling up and left. T2 is pulling up and right. The angle with the horizontal is 40 degrees on both side . See picture. What is true

 $T_1 > T_2$  $T_2 > T_1$



I missed the class

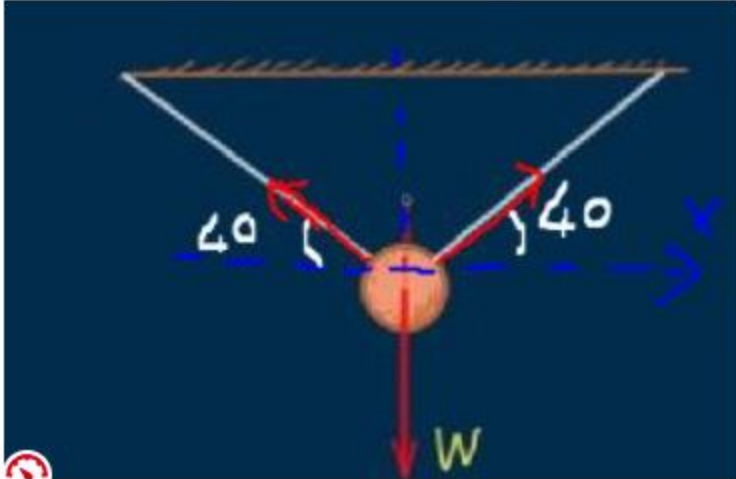


$T_1 = T_2 = T$



Question 4 15 pts

The weight is 500N. Its an equilibrium. Building on previous question. What is the right equation to write along the y-axis. (using magnitude of the components)



$T \cos(40) + T \cos(40) = 500$



$T \sin(40) + T \sin(40) = 500$ with $T_1 = T_2 = T$



$T_1 = 0, T_2 = 500\text{N}$



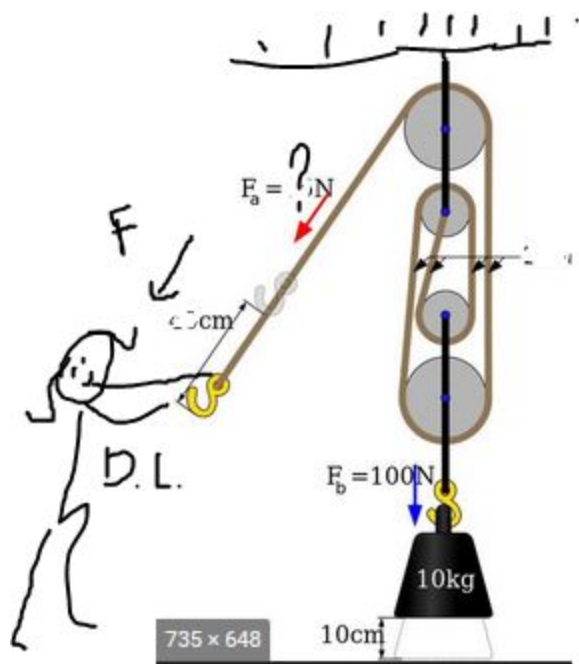
$T \sin(40) = 500$ with $T_1 = T_2 = T$



$T = 250\text{N}$ with $T_1 = T_2 = T$



Question 5 15 pts



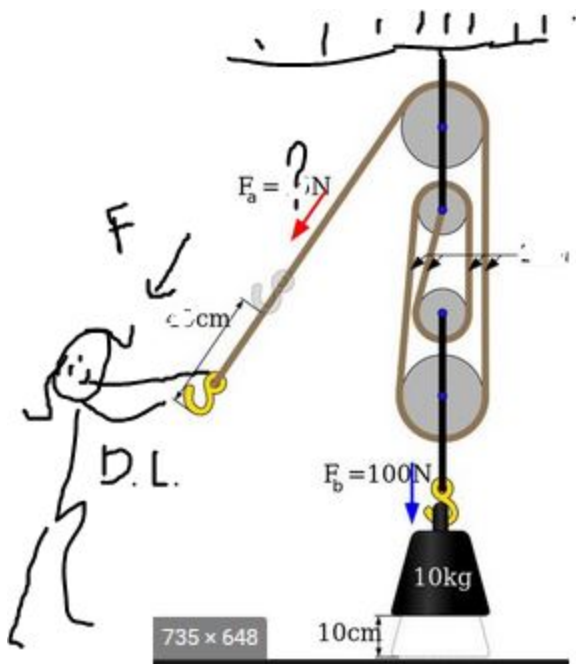
The mass is 10kg (100N weight) . I pull with a force F . The magnitude of F is

hint: Use 100N (weight) not 10kg. also place the two bottom pulleys in a "bag" . how many ropes pulling the bag you see?

- ☐ 50N
- ☐ 100N
- ☐ 25N
- ☐ I missed the class



Question 6 15 pts



If I move the weight by 10cm, I have to pull

hint: no free lunch. $\text{DISTANCE} \times \text{effort} = \text{LOAD} \times \text{distance}$

☐

10cm

☐

I missed the class

☐

40cm

☐

20cm

⋮

Question 7 15 pts

Go to :

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html

click everything that can be clicked. Place another 50kg crate on top of the previous one.

Place the mouse of the dummy and have it push the crates. go slowly. What is the maximum static friction? and the coefficient?

Make sure its 2 crates. not just one

☐

about 300N, 0.3

☐

about 200N, 0.2

☐

about 125N, 0.125

☐

about 250N, 0.25

☐

about 250, 25



Question 8 10 pts

building on the previous question. Whats the coefficient of kinetic friction

☐

about 0.2

☐

about 0.1

☐

about 0.5

☐

about 0.3

Not saved

Submit Quiz